

Decision Letter for Submission #878

Many thanks for your submission. The topic is timely and the paper is great fit with the aims and scope of Meta-Psychology. I asked two experts on meta-analysis (Joseph Hilgard and Moritz Heene) and one expert on moral psychology (Jens Agerström,) to review it and to my delight they all agreed to review it. I also announced it on Twitter and on Facebook, but no one contributed with Open Peer Review. I considered whether to invite the authors of the previous meta-analysis but decided against it. This paper isn't a commentary on their work but a clear incremental contribution building upon it.

All three reviewers are very positive toward your manuscript. They also unanimously agreed that it needed some revision before it could be accepted for publication. I agree. I very much enjoyed reading your manuscript, but there's room for expanding and clarifying certain issues. That's why I've reached the decision of Minor revision.

Hilgard and Heene provide tremendously helpful suggestions for how to strengthen the statistical part of your paper; both in terms of argument (e.g., why Fail-safe N is indeed a Fail) and in how to improve (e.g., two step PET, PET-SEE; additional plots). Agerström focused his review on how to improve the theoretical and conceptual part of your manuscript, that I agree is a little bit weak, especially for people who are new to this research area. I know it's a bit of a cliché for an editor to ask authors to carefully address *all* of the concerns raised by the reviewers, but in this case the reviewer comments are all very helpful, all point very much in the same direction, and would likely greatly improve the manuscript.

When revising your manuscript, keep in mind that we don't have any strict word limit, or limits on number of figures as other journals do. Keep this in mind when addressing the comments that generally call for expansion. That said, I enjoyed the brevity and focus of your manuscript and you should try keeping that intact.

After the reviews, I have attached comments from my own initial reading. As you will see, they echo some of the concern of the reviewers, but they might still be helpful.

Looking very much forward to reading your revised version. When you upload it, make sure to also update your preprint.

Best,
Rickard Carlsson
Editor-in-Chief
Meta-Psychology

Joseph Hilgard:

The authors inspect the moral licensing literature for signs of publication bias, using more powerful tools than previous applications of trim-and-fill. Strengths of the manuscript include use of 3PSM and the inspection and correction of previous meta-analytic data. I don't know much about this particular literature so I will focus on the bias adjustment.

I would appreciate greater detail about how multiple effect sizes within a study were handled. In the table, you report that some effect sizes were averaged together. It could be useful to state whether this was generally your approach to multiple effects, and it may also be useful to have a data set from before the averaging, giving each independent cluster a unique ID.

This is relevant because, in the case that multiple outcomes are extracted from a study and averaged together, the performance of 3PSM may suffer. This is because 3PSM models the probability of a study's publication as a single outcome, rather than an average of outcomes. Unfortunately, there is not currently any single best practice. In introducing p-curve, Simonsohn recommends extracting the single most important outcome from each study. I have experimented with bootstrapping, selecting one effect at random from each cluster, then bootstrapping a thousand times. This represents the uncertainty in choice of effect size from the study, which seems reasonable to me, but is not yet canon. You could just warn your reader about the possible error introduced by averaging and using 3PSM.

Please confirm that the individual effect sizes are completely independent. I am confused by the crop of studies credited to Simbrunner & Schlegelmilch (2016a) which make up the entire south-east Asian sample of studies. All of these studies have sample size $N = 111$, which suggests to me that they may not be independent effect sizes, but perhaps multiple effect sizes from the same study. A similar concern applies to the set of 5 effect sizes from Simbrunner & Schlegelmilch (2016a) with sample size $N = 57$. Please confirm the origin of these effect sizes and that they are fully independent. If they are not independent, please take appropriate measures (averaging, clustering, multi-level modeling). I recognize that this may be an error on S&S's part carried forward to this analysis and that investigating this will likely require correspondence from S&S.

The "Conclusion" section is a little too firm given that the manuscript does find evidence for significant moderation even after adjustment for bias and a significant effect among the North American subset. Perhaps it would also be useful and interesting to mention briefly how these estimates differ from those of previous meta-analyses on the same topic.

I'd appreciate some diagnostic plots, e.g. a funnel plot, or that density plot that the `weightr` shiny app makes. Figure 1 is appreciated but I generally prefer the output of `metafor::funnel()`, especially if one can add significance lanes. See my additions to your analysis script.

Moritz Heene

This is a very interesting manuscript which might help clarify the confusions surrounding the claimed existence of moral licensing.

However, I made some observations which might help improve the manuscript.

p. 2: "...the adjusted effect sizes were reduced to $d = -.10$ and $d = .10$, respectively." Please also report the associated p-values.

p. 3: "Fail-safe N, on the other hand, suffers from at least two issues: (1)

QRPs that increase the type-1-error rate artificially increase the estimated fail-safe N (Simonsohn, Nelson, & Simmons, 2014) and (2) fail-safe N is merely concerned with statistical significance and not effect sizes.(...) The use of fail-safe N has thus been discouraged in favor of other available methods (Becker, 2005)."

- The Fail-safe N (FSN) suffers from a more fundamental problem, an embarrassing computation error which went largely unheeded. This is because the FSN treats the file drawer of unpublished studies as *unbiased* by assuming that their average Z-value is zero. But if only 5% of studies that show Type I errors were published, the mean Z-value of the remaining unpublished studies cannot be zero but must be negative (see Scargle, 2000, p. 5, equation (3)). To illustrate this, run the following R code:

```
z <- rnorm(100000,0,1) # z-values from  $Z \sim N(0,1)$ 
z.trunc <- z[z<=qnorm(.95)]
```

```
# Truncated distribution: 95% of the studies that have 5% or
greater chance of being statistical fluctuations.
mean(z.trunc)
```

This yields:

```
> z <- rnorm(100000,0,1)
> z.trunc <- z[z<=qnorm(.95)]
> mean(z.trunc)
[1] -0.1058414
```

Consequently, the FSN is a gross overestimate of the number of unpublished studies required to bring the mean Z-value of published studies to an insignificant level. To further illustrate this point, consider the critique of Darlington (see Schonemann & Scargle, 2008, p. 22), referring to an example by Rosenthal and Rubin (1978): „...imagine that all the test null hypotheses in a certain area are true, and that the only results published are 5% of studies which achieve significance by chance. Suppose that 345 were published this way. (...) Then we are imagining that the total number of studies performed was $T = 20 \cdot 345$ or 6900' – not 65, 467, as Rosenthal and Rubin had reported.“ Based on these results, Scargle (2000, p. 102) therefore concluded: „The widely used fail-safe file-drawer (FSFD) analysis is irrelevant because it treats the inherently biased file drawer as unbiased and gives grossly wrong estimates of the size of the file drawer.“ I therefore think it is worth citing Scargle (2000) and Scargle and Schonemann (2008) to provide a more complete account of the critique on the FSN.

Concerning the PET-PEESE analysis as conducted and reported on p. 10. I suggest the following:

PET is a linear model assuming that the publication bias is constant at all levels of precision. This assumption is fulfilled when the null hypothesis is true but wrong when the H_0 is false. In this case, all the small-sample studies will suffer from file-drawer problems in contrast to big-sample studies which will be well-powered without file-drawer problems. In this case, PET tends to underestimate true effects.

In contrast, PEESE applies a quadratic model and thus makes the small-study effect weaker for those studies located at the top of the funnel plot. This is a correct assumption when the H_0 is false, but is incorrect when the H_0 is true

Given these issues, I strongly suggest following Stanley's (2017) recommendation of trying a conditional PET-PEESE procedure to tackle the two weaknesses. First, run PET. If this analysis yields a non-significant result, retain H_0 and stick with PET. If PET yields a significant result, reject H_0 and use to PEESE.

The following may just be a picky detail which, on the other hand, might improve the overall validity of the moderator analysis.

On p. 11 the authors report results of a moderator analysis with the following predictors: type of comparison, culture, and studies' standard errors. This analysis does not account for possible interaction effects among these predictors. This can be accomplished by running a moderator analysis using classification and regression trees (Xinru, Dusseldorp, Liu, & Meulman, 2017; see also <https://cran.r-project.org/web/packages/metacart/vignettes/metacart-vignette.pdf>). I ran this kind of analysis and found no support for possible interaction effects, only for culture (world_region): The R code can be found in the Appendix of this review (see below). Here are the results:

```
> set.seed(43768)
> random.effects.mod <- REMrt(formula = yi ~ comparison +
world_region + se, vi=vi, data = dat, c = .5)
> summary(random.effects.mod)

Random Effects meta-tree (K = 97 studies);
REMrt(formula = yi ~ comparison + world_region + se, data = dat,
vi = vi, c = 0.5)

A tree with 2 terminal nodes was detected
Moderators were detected as: world_region

Test for Between-Subgroups Heterogeneity under RE assumption:
Qb = 24.503 (df = 1), p-value < 1e-04;
The estimate for the residual heterogeneity tau2 = 0.052;

Subgroup Meta-analysis Results:
K g se zval pval ci.lb ci.ub
2 5 -0.372 0.133 -2.797 0.005 -0.633 -0.111 **
3 92 0.309 0.035 8.790 0.000 0.240 0.378 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

I think this additional step would contribute to a more complete analysis overall.

Minor points:

- APA 6th style: Greek letters should not be italicized.
- Please provide citations of the R packages you used for data analysis. This applies to the weightr package which has no citation.

References

Scargle, J. D. (2000). Publication Bias: The “File-Drawer” Problem in Scientific Inference. *Journal of Scientific Exploration*, 14, 91–106. Retrieved from <https://pdfs.semanticscholar.org/ad26/5c494be00224f1d10a14684b2dce331efc01.pdf>

Schonemann, P. H., & Scargle, J. D. (2008). A Generalized Publication Bias Model. *Chinese Journal of Psychology*, 50, 21–29. Retrieved from <https://arxiv.org/ftp/arxiv/papers/0808/0808.1588.pdf>

Stanley, T. D. (2017). Limitations of PET-PEESE and Other Meta-Analysis Methods, Limitations of PET-PEESE and Other Meta-Analysis Methods. *Social Psychological and Personality Science*, 8(5), 581–591. <https://doi.org/10.1177/1948550617693062>

Xinru Li, Elise Dusseldorp, Kaihua Liu and Jacqueline Meulman (2017). metacart: Meta-CART: A Flexible Approach to Identify Moderators in Meta-Analysis. R package version 1.0-0. <https://CRAN.R-project.org/package=metacart>

Appendix:

```
# Moderator analysis using classification and regression trees

library(metacart) # A Flexible Approach to Identify Moderators in
Meta-Analysis

# load corrected Simbrunner & Schlegelmilch dataframe
dat <- read.table("dat_new_s.txt", sep=" ", header=T)

# define variables
dat$comparison <- ifelse(dat$comparison=="1","neutral","immoral")
dat$se <- as.numeric(as.character(dat$se))
dat$yi <- as.numeric(as.character(dat$yi))
dat$vi <- dat$se^2

# reassign levels
dat$world_region <- factor(dat$world_region,levels=c("SEA","NOA","EUR"))

# Moderator analysis
set.seed(43768)
random.effects.mod <- REMrt(formula = yi ~ comparison + world_re-
gion + se, vi=vi, data = dat, c = .5)
summary(random.effects.mod)
plot(random.effects.mod)
```

Output:

```
> set.seed(43768)
> random.effects.mod <- REMrt(formula = yi ~ comparison +
world_region + se, vi=vi, data = dat, c = 1)
```

```
> summary(random.effects.mod)
```

```
Random Effects meta-tree (K = 97 studies);
REmrt(formula = yi ~ comparison + world_region + se, data = dat,
vi = vi, c = 1)
```

```
A tree with 2 terminal nodes was detected
Moderators were detected as: world_region
```

```
Test for Between-Subgroups Heterogeneity under RE assumption:
Qb = 24.503 (df = 1), p-value < 1e-04;
The estimate for the residual heterogeneity tau2 = 0.052;
```

```
Subgroup Meta-analysis Results:
```

```
K g se zval pval ci.lb ci.ub
2 5 -0.372 0.133 -2.797 0.005 -0.633 -0.111 **
3 92 0.309 0.035 8.790 0.000 0.240 0.378 ***
```

```
---
```

```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Jens Agerström

Having a keen interest in moral psychology, I read this well written paper with excitement. The current paper is to a large extent a re-analysis of the most recent and comprehensive data set (Simbrunner & Schlegelmilch, 2017) on moral-licensing. Moral-licensing as such has received a lot of attention within the field of moral psychology and within social psychology more generally. Being at odds with several other major social psychological theories (e.g. various consistency theories), it has sparked substantial controversy. Thus, it is not surprising that previous replication studies exist on the topic (e.g., many-labs replication project). The authors argue that if one corrects for publication bias using better methods (PET-PEESE and the 3-PSM) and use a larger data set that also includes unpublished studies, this result in substantially lower moral licensing effect size estimates that come close to zero ($d = .10$, and $d = .10$). My view is that the paper does make an (incremental) contribution to the meta-psychological literature on moral licensing. Besides, using a more inclusive data set, regardless of the superiority of PET-PEESE and the 3-PSM (which my expertise is too limited to assess), simply showing that using different methods to account for publication bias can render considerable effect size differences seems important. Also, meticulously comparing and looking for coding discrepancies between the meta-analyses of Blanken et al. (2015) and Simbrunner and Schlegelmilch (2017) is praiseworthy and a nontrivial metapsychological task.

Below, I offer some recommendations for how the paper can be improved. Most of these improvements are conceptual in nature and concern moral licensing as a phenomenon, which has received little appreciation by the authors whose focus has mainly been methodological/statistical.

1. When reading the theoretical background, it struck me that the mechanisms theorized as driving the moral licensing effect are only vaguely described. Moral licensing as such is defined in the paper but the psychological processes underlying it are not thoroughly mentioned. A concrete example of this would also help to help readers not familiar with the concept and underlying theory. The authors' meta-psychological focus probably contributes to this lack of description. However, the reader should not have to consult the original work (e.g., Monin & Miller, 2001) to obtain a satisfactory understanding of the underlying theory.
2. Culture is mentioned in the introduction as a potential moderator through its connection to self-concept. The reader would be helped if this aspect is elaborated. That is, how does culture influence self-concept and subsequently moderate moral licensing? This seems especially relevant given the authors' finding that culture is a significant moderator even after adjusting for publication bias. Although ego-depletion is conceptually different, the authors' finding that the moral licensing effect was reversed in the south east Asian sample, with these participants displaying an increase in moral behavior after having previously acted morally, I cannot help thinking about the reversed ego-depletion effect found when studying Indian samples (see Savani & Job, 2017, in JPSP).
3. The mentioning of self-concept in relation to culture raises the question of self-concept as an individual difference variable that I suspect would act as a moderating variable more generally. I cannot see how one could conduct moral licensing research without taking (moral) self-concept, or values, into consideration. After all, moral **self**-licensing or moral **self**-regulation were the terms originally used (see e.g., Merritt, Efron, & Monin, 2010; Sachdeva, S., Iliev, R., & Medin, D. L. (2009). Would inherently prosocial people (e.g., Mother Theresa) be expected to show the effect? What about psychopaths? This is related to my first point stressing that the paper would benefit from including a more thorough description of the psychological processes theorized to underlie the moral licensing effect.
4. Methodological issue. Because the current authors argue that publication bias contributes to the moral licensing effect size being overestimated in previous meta-analyses such as that of Simbrunner and Schlegelmilch (2017), I cannot help to wonder to what extent also truly failed manipulations account for research with null findings not being published on this topic. Would not publishing such failed studies also be considered as publication bias by the authors? Have the previous meta-analyses taken this into account? Given that the experimental manipulations in the context of moral licensing do not seem to be

straightforward, and hinge on moral values among other things, this seems particularly important to consider when conducting a meta-analysis. Because this was not done in the current meta-analysis, it should minimally be mentioned as a limitation.

5.

3-PSM. The logic on which 3-PSM builds needs to be explained in more detail.

6.

The authors state that an effect size of $d = .10$ has no practical importance. In my view, effect sizes have to be evaluated in context. See, for example, Greenwald, Banaji, & Nosek's (2015) JPSP-paper and Messick's (1995) original paper on consequential validity. If the authors think that a small statistical effect of moral licensing cannot be translated into any meaningful societal effect, I suggest that they at least argue why they believe so.

7.

Future research. Based on their results, the authors suggest that future research (ideally preregistered, highly powered studies) examine culture as a moderating variable. Again, as a result of not discussing the link between culture, self-concept/values and moral licensing, the reader is completely left to his own devices here. As such, the suggestions for future research are quite empty and the paper does not contribute much in this respect.

Rickard Carlsson:

- Why no confidence intervals? If you have some reason to dislike confidence intervals, how about reporting SE along with z?
- I-squared is sometimes hard to interpret. I suggest report and interpret tau.
- Figure 1 looks quite heteroscedastic. Is this a problem? (Might be solved by point raised by Heene on two step PET, PET-PEESE)
- Could you add figures for the subgroup analysis too? Might help tease out the publication bias vs. culture thing.
- Related to culture, and point raised by Agerström (as well as the interaction point raised by Heene) is it possible that the effect is reversed in other cultures because these are later papers and publication bias favors ironic effects over null effects? Lets' say the effect is weaker, and actually zero, in this group, but people from other cultures are able to "sell" these papers by making a reversed effect "sexy"? A simple reading of the papers narrative might reveal if there's anything to this point.
- What is the heterogeneity level within the culture subgroups? The PET-PEESE is quite different there for NA and amount of heterogeneity might be relevant then.
- Personally, I think contour enhanced funnel plots are very helpful when interpreting meta-analysis. (also see comment by Hilgard)
- Bias estimation / correction should ideally be done on the test statistic that was potentially censored / hacked. I would like to know more about whether this is true for the cases where there is more than one outcome. One way to go about this is to do an alternative analysis based on focal stats extraction and estimate indices such as *P*-curve, R-index and Z-curve (currently under review at MP so you can find the code on our OSF project hub). This is just a suggestion – see similar comment by Hilgard.